E-Commerce Sales and Profit Analytics Using Data Engineering Pipeline

1st Jovita Niken Angelia Putri
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia
jovita.putri@binus.ac.id

2nd Felix
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia
felix023@binus.ac.id

3rd Filbert Tanuarta
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia
filbert.tanuarta@binus.ac.id

4th Richard Vinson
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia
richard.vinson@binus.ac.id

5th N.G. Christian Nababan
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia
ng.nababan@binus.ac.id

6th Yusuf Rayyan Daniel
Computer Science Department
Bina Nusantara University
Jakarta, Indonesia
yusuf.daniel@binus.ac.id

***Abstract*—This project creates and sets up a full data engineering system to analyze e-commerce sales and profits. The pipeline uses an e-commerce transactional dataset as its main source of data and goes through different steps in the data engineering process.These steps include getting the data in, storing it, changing it into the right format, creating models for the data, checking its quality, setting up automation, and finally using the data for its intended purpose. Pentaho Data Integration helps take the raw data and turn it into a clear, organized format that's easier to analyze. The data that has been changed is then arranged into a star schema, which includes fact tables and dimension tables, to help with faster reporting and analysis. Data quality checks are done to deal with missing information, different formats, repeated entries, and wrong business data. The final processed dataset is used in a Power BI dashboard that shows insights about sales, profit, products, customers, and different regions. This project shows how organizing data through a clear data engineering process can make raw transaction data more dependable and easier to use for making business decisions.**

Index Terms *— Data Engineering, ETL, Pentaho Data Integration, E-Commerce Analytics, Data Warehouse, Star Schema, Data Quality, Power BI, Business Intelligence.*

I. INTRODUCTION

E-commerce has grown very fast in recent years, and this growth has created a huge amount of data. This data comes from many sources, such as customer orders, product sales, deliveries, regional operations, and profit records. Even though this data is very useful for making business decisions, it usually cannot be used right away. Raw data often has problems like missing information, duplicate entries, different formats, and messy relationships between data. Because of this, businesses need a clear and organized process to turn raw data into something useful and trustworthy.

Data engineering helps solve this problem. With an ETL process which stands for Extract, Transform, and Load data can be taken from its original source, cleaned up, checked for errors, and saved in a neat structure. Using a data model like a star schema also helps a lot. It organizes data into fact tables and dimension tables, which makes it easier to create reports and do business analysis. When data is properly cleaned and well-organized, it becomes more accurate and ready to support good decision-making.

This project builds a complete data engineering pipeline for e-commerce sales and profit analysis. The pipeline goes through many steps, including finding the data source, loading data using Pentaho Data Integration, storing raw and processed data, transforming the data, building an analytical data model, checking data quality, automating the process, and preparing the data for use. The final output supports a Power BI dashboard that shows insights about sales, profit, products, customers, and regional performance. This project shows how data engineering can turn messy raw data into clean, structured data that helps businesses make better decisions.

II. PROJECT OVERVIEW

This project is about building a complete data engineering pipeline for e-commerce sales and profit analysis. The main goal is to take raw e-commerce data and turn it into clean, well-organized data that is ready to use for business reporting. The data includes information about sales transactions, products, customers, shipping, regional performance, and profit.

The pipeline follows the main steps of the data engineering lifecycle. First, the data source is identified and understood. Then, the data is loaded using Pentaho Data Integration. After that, the raw data is saved in a storage environment and goes through several transformation steps, such as cleaning, checking for errors, fixing formats, joining tables, and summarizing data. Once the data is transformed, it is organized into an analytical model using a star schema, which is made up of fact tables and dimension tables.

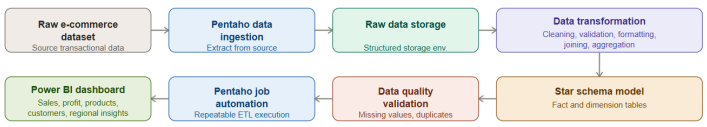


Fig. 1. Overview of the e-commerce data engineering pipeline.

To make sure the data is reliable, quality checks are done to find and fix problems like missing values, duplicate records, wrong formats, and invalid data. The ETL workflow is also automated using Pentaho Jobs, so the pipeline can run in a consistent and repeatable way. The final result of this project is a Power BI dashboard that uses the processed data to show useful insights about sales performance, profit trends, product categories, customer behavior, and regional performance.

In general, this project shows how data engineering can turn raw e-commerce data into reliable analytical data for dashboard reporting. By combining ETL processing, structured storage, star schema modeling, data quality checks, automation, and Power BI visualization, the pipeline makes data more useful and easier to analyze for business decision-making.

III. METHODOLOGY

A. Data Source and Data Generation

The dataset used in this project is the Sample Superstore dataset, which comes in CSV format. It contains sales transaction data from a retail business, including information about orders, customers, products, shipping, regions, sales amounts, discounts, quantities, and profit. The dataset has 10,194 records and 21 attributes, which makes it a good fit for building a data engineering pipeline and supporting business analysis through a dashboard.

The main focus of this dataset is sales and profit analytics. Each record represents one sales transaction that can be looked at from different business angles, such as customer segment, product category, shipping mode, region, sales value, and profit contribution. Because the dataset has both transactional and descriptive fields, it can be organized into fact and dimension tables for analytical reporting.

TABLE I

DATASET OVERVIEW

Description	**Information**
Dataset Name	Sample Superstore Dataset
File Format	CSV
Number of Records	10,194 rows
Number of Attributes	21 columns
Data Context	Retail / e-commerce sales transaction
Main Measures	Sales, Quantity, Discount, Profit
Main Dimensions	Customer, Product, Date, Region, Shipping
Output Use	Power BI dashboard

Order ID	Order Date	Ship Mode	Customer ID	Segment	Region	Category	Sub-Category	Sales	Quantity	Discount	Profit
US-2023-103800	1/3/2023	Standard Class	DP-13000	Consumer	Central	Office Supplies	Paper	16.45	2	0.2	5.55
US-2023-112326	1/4/2023	Standard Class	PO-19195	Home Office	Central	Office Supplies	Binders	3.54	2	0.8	-5.49
US-2023-112326	1/4/2023	Standard Class	PO-19195	Home Office	Central	Office Supplies	Labels	11.78	3	0.2	4.27
US-2023-112326	1/4/2023	Standard Class	PO-19195	Home Office	Central	Office Supplies	Storage	272.74	3	0.2	-64.77
US-2023-141817	1/5/2023	Standard Class	MB-18085	Consumer	East	Office Supplies	Art	19.54	3	0.2	4.88

Fig. 2. Preview of the raw Sample Superstore dataset.

TABLE II

ATTRIBUTE DESCRIPTION

Attribute	**Description**
Row ID	Unique row number for each record in the dataset
Order ID	Unique identifier for each customer order
Order Date	Date when the order was placed
Ship Date	Date when the order was shipped
Ship Mode	Shipping method used for the order
Customer ID	Unique identifier for each customer
Customer Name	Name of the customer
Segment	Customer segment, such as Consumer, Corporate, or Home Office
Country/Region	Country or region where the transaction occurred
City	City of the customer or transaction location
State/Province	State or province of the customer or transaction location
Postal Code	Postal code of the transaction location
Region	Sales region, such as Central, East, South, or West
Product ID	Unique identifier for each product
Category	Main product category
Sub-Category	More specific product category
Product Name	Full name of the product sold
Sales	Sales value generated from the transaction
Quantity	Number of products purchased

Discount	Discount applied to the transaction
Profit	Profit generated from the transaction

B. Data Ingestion

The data ingestion process in this project is carried out using Pentaho Data Integration. The raw dataset, which is saved in CSV format, is taken from the source file and brought into the data processing environment. This step is important because it is the starting point for moving the raw e-commerce data into the pipeline before any cleaning, transformation, validation, or modeling is done.

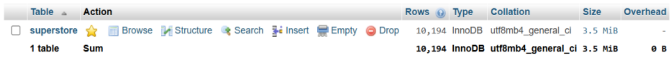


Fig. 3. Ingestion.

A full load strategy is used in this project because the Sample Superstore dataset is handled as one complete batch of historical data. Instead of only loading new or changed records, the entire CSV file is extracted and loaded into the system at once. This approach works well for this project because the dataset is static and does not need real-time or incremental updates.

In Pentaho, the ingestion process starts by reading the CSV file through a mysql. The fields in the dataset, such as order details, customer information, product data, sales, quantity, discount, profit, and regional information, are matched according to their data types. Once the data is read successfully, it is loaded into a raw storage or staging area. This raw storage keeps the original structure of the dataset before any transformation is applied, so the data can still be checked and compared with the processed output later if needed.

In short, the data ingestion stage makes sure that the raw Sample Superstore dataset is properly brought into the pipeline and ready to be used in the next stages.

C. Data Storage

The data storage stage is set up to keep both raw and processed data in an organized way. In this project, the raw Sample Superstore dataset is first saved as the original input before any changes are made. Keeping this raw storage is important because it holds the original structure of the dataset and can be used as a backup in case something goes wrong during the transformation process.

After the ingestion stage, the data is processed and moved into a separate storage area for transformed data. This processed storage holds data that has already been cleaned, validated, formatted, and prepared for analysis. Separating the raw and processed data makes it easier to track changes, since the original data and the transformed output can be compared whenever needed.

The raw data storage contains the full transactional dataset from the CSV file, including order details, customer information, product data, shipping records, sales figures, discounts, quantities, profit values, and regional information. The processed data storage, on the other hand, contains structured tables that are used for analytics and reporting. These tables include dimension tables and a fact table, which are used later to support the star schema model and the Power BI dashboard.

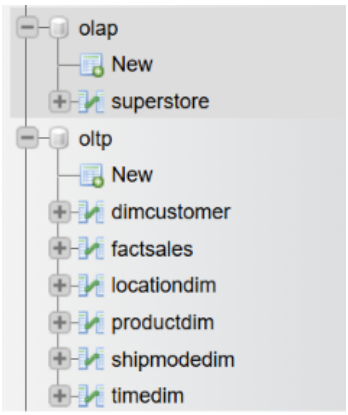


Fig. 4. Raw and New.

SBy dividing the data into raw and processed storage, the pipeline becomes more reliable and easier to manage. The raw storage keeps the original data untouched, while the processed storage provides clean and well-structured data that is ready for business intelligence reporting.

D. Data Transformation

The data transformation stage is done to turn the raw Sample Superstore dataset into clean, consistent, and analysis-ready data. This stage is important because raw transactional data often has problems like inconsistent formats, duplicate records, missing values, or fields that are not yet ready for dashboard reporting. In this project, the transformation process is carried out using Pentaho Data Integration before the data is loaded into the analytical data model.

Several transformation steps are applied in this stage. First, unnecessary or duplicate records are checked to make sure repeated transactions do not affect the analysis. Second, data formats are standardized, especially for date fields, numerical fields, and categorical fields. Date attributes like Order Date and Ship Date are formatted in a consistent way, while numerical attributes like Sales, Quantity, Discount, and Profit are prepared for aggregation and calculation.

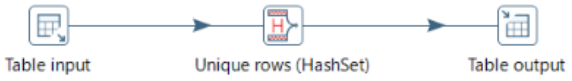


Fig. 5. Transformation.

The transformation process also includes selecting and preparing the right fields for analytical use. Transactional measures such as Sales, Quantity, Discount, and Profit are set up as measurable values for the fact table. Descriptive attributes such as customer information, product category, location, shipping mode, and order date are prepared for the dimension tables. This separation allows the data to be organized into a star schema model in the next stage.

Aggregation is also done to support business analysis in Power BI. The processed data can be grouped and summarized by product category, customer segment, region, order date, and shipping mode. These summaries help produce key business metrics such as total sales, total profit, total quantity sold, average discount, and profit contribution by category or region.

Overall, the data transformation stage makes sure that the raw dataset is turned into structured and reliable data. The transformed output is then used as the foundation for data modeling, data quality validation, and Power BI dashboard visualization.

E. Data Modeling

The data modeling stage is done to organize the transformed Sample Superstore data into an analytical structure that supports reporting and business analysis. In this project, a star schema approach is used because it separates measurable transactional data from descriptive business attributes. This structure makes the data easier to analyze in Power BI and supports different reporting perspectives, such as sales by product category, profit by region, and customer performance by segment.

The star schema has one main fact table and several dimension tables. The fact table stores quantitative business values, including Sales, Quantity, Discount, and Profit. These values are used to calculate key performance indicators in the Power BI dashboard. The dimension tables store descriptive information related to customers, products, locations, order dates, and shipping. These dimensions allow users to filter, group, and compare business performance from different angles.



Fig. 6. Star Schema.

By using a star schema, the processed data becomes more structured, consistent, and ready for business intelligence reporting. This model also makes the dataset easier to use because Power BI can work with the fact and dimension tables to create interactive visuals, filters, and business performance summaries.

F. Data Quality and Validation

The data quality and validation stage is carried out to make sure the transformed dataset is accurate, complete, and ready for analytical reporting. Because the quality of the data has a direct effect on how reliable the business insights are, validation steps are done before the data is loaded into the analytical model and Power BI dashboard.

The validation process covers three main areas: missing values, duplicate records, and data consistency. Key fields such as Order ID, Customer ID, Product ID, Sales, Quantity, and Profit are reviewed to make sure no important information is missing. Duplicate records are also found and flagged to prevent wrong calculations of sales and profit figures.

Besides that, data type validation is applied to make sure each field is in the right format. Date fields like Order Date and Ship Date are checked to confirm they are stored as proper date values. Sales, Quantity, Discount, and Profit are checked to make sure they are in numerical format. Categorical fields such as Segment, Category, and Region are also reviewed to keep the data consistent throughout the dataset.

By doing these validation steps, the dataset becomes more reliable and ready for business intelligence analysis. The quality checks help lower the chance of errors and make sure the Power BI dashboard gives accurate insights about sales performance, profit trends, customer behavior, and regional business performance.

G. Automation and Orchestration

The automation and orchestration stage is set up to make sure all ETL processes can be run in a structured and repeatable way. In this project, Pentaho Jobs are used to manage the execution of the data engineering pipeline, which includes data ingestion, transformation, validation, and data modeling.

The workflow is arranged in a specific order so that each step runs at the right time. The pipeline starts by loading the raw Sample Superstore dataset, then moves on to the data transformation and validation steps. Once the data has been cleaned and validated, the pipeline continues with creating the dimension tables and the fact table that make up the analytical data model.



Fig. 7. ETL Workflow Automation.

By using Pentaho Jobs, the entire workflow can be controlled from one place. This reduces the need for manual work, keeps the process consistent, and makes sure every step in the pipeline is completed successfully before the data is used in Power BI. As a result, the pipeline becomes more reliable and easier to maintain for future reporting needs.

H. Data Consumption

The last stage of the data engineering pipeline is data consumption. In this project, the processed analytical dataset is imported into Power BI to build interactive dashboards and business intelligence reports. Power BI is used as the main platform for showing business insights that come from the transformed and modeled data.

The dashboard displays key business metrics such as total sales, total profit, total quantity sold, and average discount. Several visuals are also created to look at sales performance by product category, customer segment, shipping mode, and

geographical region. Time-based analysis is included as well to help identify sales and profit trends over different time periods.

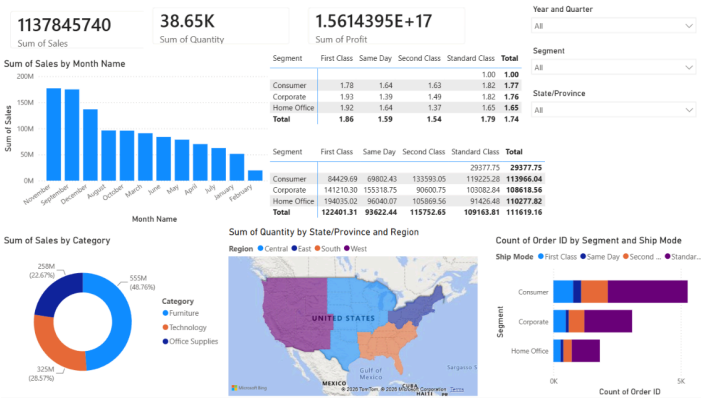


Fig. 8. Power BI Dashboard.

The dashboard lets users explore business performance through interactive filters and visuals. Users can compare sales and profit across different categories, find high-performing products, check regional performance, and look at customer purchasing patterns. In this way, the dashboard helps users monitor business performance more effectively and make better decisions by turning processed data into clear and useful insights.

IV. RESULTS AND DISCUSSION

A. Dashboard Results

The processed dataset was successfully brought into Power BI to build an interactive dashboard for sales and profit analysis. The dashboard shows key performance indicators, including Total Sales, Total Profit, Total Quantity Sold, and Average Discount. Several visuals were also created to look at business performance based on product categories, customer segments, regions, and time periods.

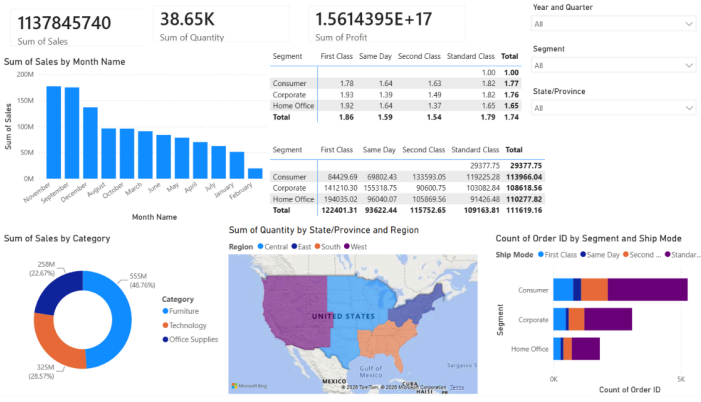


Fig. 9. Power BI Dashboard.

The dashboard allows users to explore business performance through interactive filters and visual displays. By using the star schema model that was built during the data modeling stage, the dashboard can support multidimensional analysis and reporting in an efficient way.

B. Business Insights

Analysis of the executed Power BI dashboard shows important, actionable findings about how promotion affect profits:

- Seasonal Sales Peaks and Promotional Timing: The "Sum of Sales by Month Name" chart illustrates distinct revenue spikes during the latter half of the year, most notably in November, September, and December. This indicates that end-of-year promotional campaigns successfully drive high sales volume. However, to maximize profitability, the discount rates applied during these peak months must be carefully monitored to ensure they do not excessively erode net profit margins.
- Category-Specific Discount Strategies: As shown in the "Sum of Sales by Category" visual, the Technology segment commands the largest revenue share, followed by Furniture and Office Supplies. To optimize profits, promotional budgets should be dynamically allocated. For instance, aggressive promotions on high-performing Technology products might lead to unnecessary profit loss, whereas targeted discount campaigns for Office Supplies could effectively stimulate slow-moving inventory.
- Segment and Shipping Optimization: The data regarding "Count of Order ID by Segment and Ship Mode" reveals that the Consumer segment generates the highest volume of transactions, predominantly utilizing 'Standard Class' shipping. Introducing tiered promotional offers—such as discounted 'First Class' or 'Same Day' shipping for consumers who meet a specific spending threshold—could incentivize higher average order values and improve overall transaction profitability.

C. Discussion

The implementation of the interactive Power BI dashboard highlights the critical role of effective data modeling in extracting actionable business intelligence. As demonstrated in the results, utilizing a star schema architecture significantly enhances the efficiency of multidimensional analysis. This structured approach allows stakeholders to seamlessly filter and cross-reference datasets across various dimensions, such as time periods, geographical regions, and customer segments without compromising query performance.

Furthermore, the insights derived from the visual displays underscore the necessity of transitioning from intuitive to data-driven promotional strategies. By clearly visualizing sales peaks, category-specific performance, and shipping distributions, the dashboard provides a reliable foundation for optimizing discount allocations. This ensures that promotional efforts stimulate revenue growth without causing unnecessary profit erosion, particularly during high-traffic seasonal periods.

Despite these operational advantages, the current dashboard iteration is primarily descriptive and relies on historical data batches. To further better its analytical value, future developments should focus on integrating automated data ingestion pipelines to support near real-time reporting. Additionally, incorporating predictive analytics and machine learning models could transition the system from a descriptive reporting tool into a proactive decision-support system, enabling the accurate forecasting of future sales trends and optimal discount thresholds.

V. CONCLUSION

This project successfully designed, implemented, and evaluated a complete end-to-end data engineering pipeline for

e-commerce sales and profit analysis. By executing every stage of the data engineering pipeline, including data ingestion, structured storage, data transformation, star schema modeling, data validation, automation, and consumption, the pipeline successfully converted 10,194 raw transactional records into a clean, dependable asset for business' decision making.

Through the final data consumption stage, the processed data was able to be served to an interactive Power BI dashboard. This allows for a clear, multi-dimensional view of key business metrics across product categories, customer segments, operational shipping modes, and geographic regions. The final output demonstrates that proper data engineering and structured ETL workflows are essential to turning messy, unvalidated raw data into highly accurate, trustworthy business intelligence. Future improvements to this pipeline could focus on expanding the system to handle real-time data streaming or implementing incremental loading strategies to handle growing data volumes more efficiently.

Appendix

The additional materials for this project are available on the following links:

- **Dataset: https://www.kaggle.com/datasets/himanshuuike/superstore-sales-dataset**
- **Presentation Slides: https://canva.link/ulfvq0n5mw72gju**

References

[1] H. Uike, "Superstore Sales Dataset," Kaggle. Accessed: Jun. 20, 2026 [Online]. Available: https://www.kaggle.com/datasets/himanshuuike/superstore-sales-dataset